Lecture 10

Kinematics of Rigid Bodies II

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MC2103

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Lecture 10-1

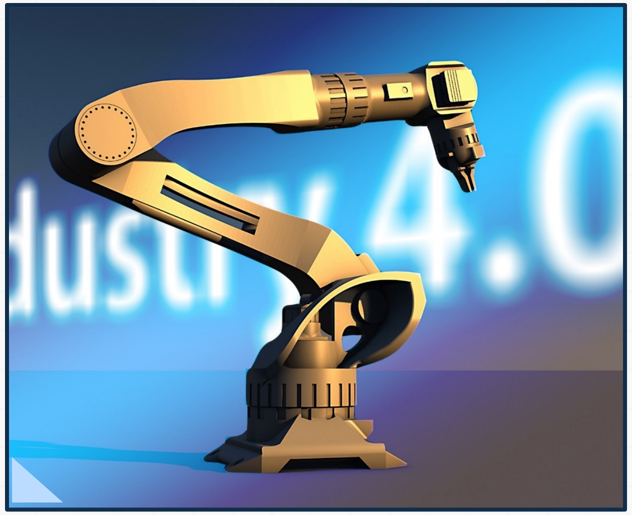
https://youtu.be/55yHlI1Adfc

Lecture 10-2

GIST

Contents

In Week 9, we studied reference frames and vector differentiation. We also studied that orientation is not a vector. In this week, we will examine more on the orientation of rigid bodies in 3D spaces.



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Orientation of Rigid Bodies

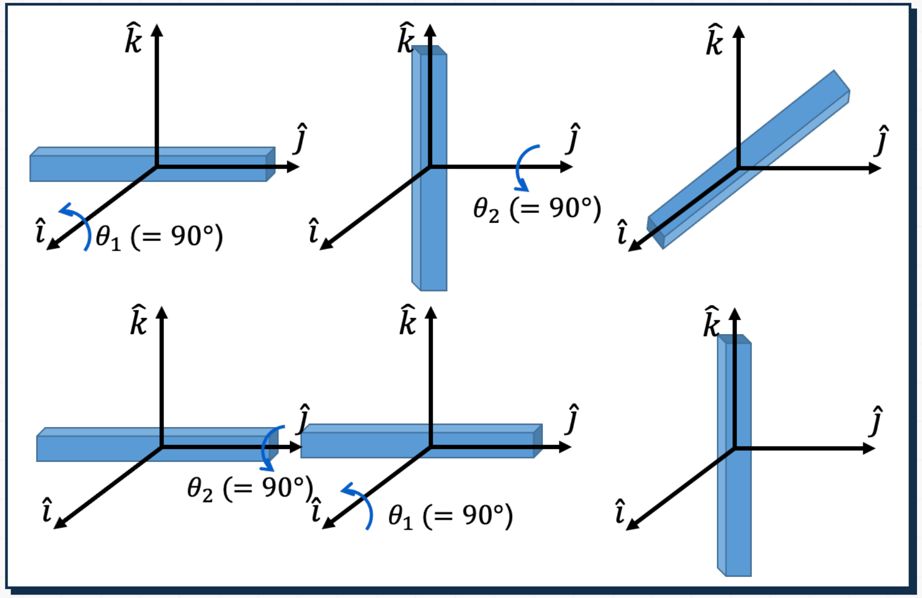
Orientation

- Orientation (or attitude) means how a rigid body is placed in 3D space.
- Orientation can be quantitatively defined with respect to a reference orientation just as a position can be quantified with respect to a reference point (i.e., an origin).

Orientation of Rigid Bodies

Orientation

- There are multiple ways of defining orientation of a rigid body.
- Also, orientation is not a vector, meaning that changing order of rotations ends up with different orientation.



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Relative Orientation

How do we represent an orientation of a rigid body with respect to a reference orientation?

- Consider a rigid body of our interest along with a body-attached coordinate system.

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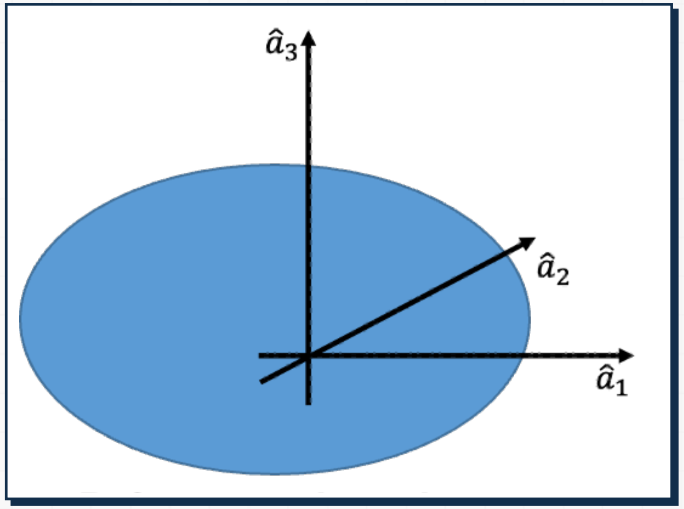
Relative Orientation

Reference Orientation



- Denote $\hat{a}_1\hat{a}_2\hat{a}_3$ the triad of the unit vectors aligned with the xyz axes of the coordinate system.
- Now, rotate the rigid body.

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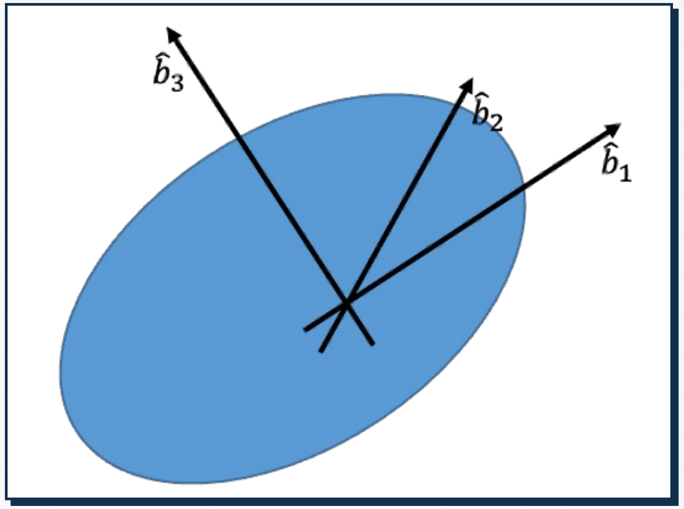
Relative Orientation

Rotated Orientation



- Denote $\hat{b}_1\hat{b}_2\hat{b}_3$ the triad of the unit vectors aligned with the xyz axes of the coordinate system for the rotated rigid.
- Direction cosine matrix maps $\hat{a}_1\hat{a}_2\hat{a}_3$ to $\hat{b}_1\hat{b}_2\hat{b}_3$.

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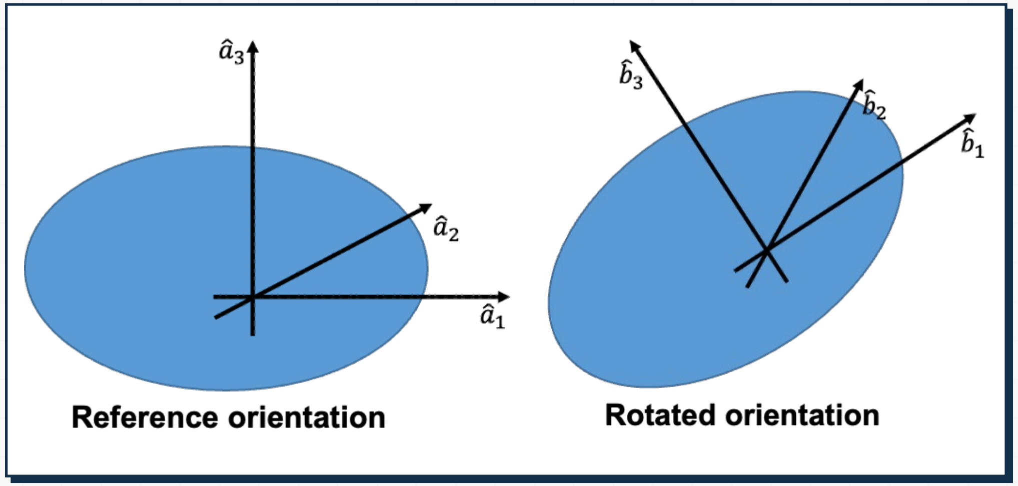
Direction Cosine Matrix

Direction cosine matrix maps $\hat{a}_1\hat{a}_2\hat{a}_3$ to $\hat{b}_1\hat{b}_2\hat{b}_3$.

- How do we find the direction cosine matrix (DCM)?



- Note that

$$\hat{a}_1 = (\hat{a}_1 \cdot \hat{b}_1)\hat{b}_1 + (\hat{a}_1 \cdot \hat{b}_2)\hat{b}_2 + (\hat{a}_1 \cdot \hat{b}_3)\hat{b}_3$$

- Similarly,

$$\hat{a}_2 = (\hat{a}_2 \cdot \hat{b}_1)\hat{b}_1 + (\hat{a}_2 \cdot \hat{b}_2)\hat{b}_2 + (\hat{a}_2 \cdot \hat{b}_3)\hat{b}_3$$
$$\hat{a}_3 = (\hat{a}_3 \cdot \hat{b}_1)\hat{b}_1 + (\hat{a}_3 \cdot \hat{b}_2)\hat{b}_2 + (\hat{a}_3 \cdot \hat{b}_3)\hat{b}_3$$

Direction Cosine Matrix

Direction cosine matrix maps $\hat{a}_1\hat{a}_2\hat{a}_3$ to $\hat{b}_1\hat{b}_2\hat{b}_3$.

- How do we find the direction cosine matrix (DCM)?
  - Putting together (with an abuse of notation),

Direction cosine matrix

$$\begin{pmatrix} \hat{a}_1 \\ \hat{a}_2 \\ \hat{a}_3 \end{pmatrix} = \underbrace{\begin{pmatrix} \hat{a}_1\cdot\hat{b}_1 & \hat{a}_1\cdot\hat{b}_2 & \hat{a}_1\cdot\hat{b}_3 \\ \hat{a}_2\cdot\hat{b}_1 & \hat{a}_2\cdot\hat{b}_2 & \hat{a}_2\cdot\hat{b}_3 \\ \hat{a}_3\cdot\hat{b}_1 & \hat{a}_3\cdot\hat{b}_2 & \hat{a}_3\cdot\hat{b}_3 \end{pmatrix}} \begin{pmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{pmatrix}$$

Direction cosine table

	$\hat{b}_1$	$\hat{b}_2$	$\hat{b}_3$
$\hat{a}_1$	$\hat{a}_1\cdot\hat{b}_1$	$\hat{a}_1\cdot\hat{b}_2$	$\hat{a}_1\cdot\hat{b}_3$
$\hat{a}_2$	$\hat{a}_2\cdot\hat{b}_1$	$\hat{a}_2\cdot\hat{b}_2$	$\hat{a}_2\cdot\hat{b}_3$
$\hat{a}_3$	$\hat{a}_3\cdot\hat{b}_1$	$\hat{a}_3\cdot\hat{b}_2$	$\hat{a}_3\cdot\hat{b}_3$

Parameterization of Orientation

Direction cosine matrix has nine elements.

- However, not every element is independent.
  - For example, each row vector in the DCM is a unit vectors(3 constraints).

$$\sum_{j=1}^{3} c_{ij}^2 = 1, (i = 1,2,3)$$

  - Also, each row vector in the DCM is orthogonal to each other(3 constraints).

$$\sum_{j=1}^{3} c_{1j}c_{2j} = 0, \sum_{j=1}^{3} c_{1j}c_{3j} = 0, \sum_{j=1}^{3} c_{2j}c_{3j} = 0$$

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Parameterization of Orientation

Direction cosine matrix has nine elements.

- Therefore, only THREE pieces of information are needed to uniquely determine the orientation. (note the following expression is an abuse of notation)

$$\begin{pmatrix} \hat{a}_1 \\ \hat{a}_2 \\ \hat{a}_3 \end{pmatrix} = \begin{pmatrix} \hat{a}_1 \cdot \hat{b}_1 & \hat{a}_1 \cdot \hat{b}_2 & \hat{a}_1 \cdot \hat{b}_3 \\ \hat{a}_2 \cdot \hat{b}_1 & \hat{a}_2 \cdot \hat{b}_2 & \hat{a}_2 \cdot \hat{b}_3 \\ \hat{a}_3 \cdot \hat{b}_1 & \hat{a}_3 \cdot \hat{b}_2 & \hat{a}_3 \cdot \hat{b}_3 \end{pmatrix} \begin{pmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix} \begin{pmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{pmatrix}$$

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Euler Angles

We learned that three pieces of information are needed to uniquely define orientation in 3D space.

- There are multiple ways of describing orientation.
- One way is using three consecutive rotations, called Euler angles.

Euler Angles

Example

- A rotation in x axis by θ, followed by a rotation in y axis by θ, then followed by a rotation in z axis by ψ.

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Euler Angles

- Rotation can also be about either body axis or space axis.

12 rotations about body axis

- XYZ, XZY, YZX, YXZ, ZXY, ZYX, XYX, YZY, XZX, ZXZ, YXY, ZYZ

12 rotations about space axis

- XYZ, XZY, YZX, YXZ, ZXY, ZYX, XYX, YZY, XZX, ZXZ, YXY, ZYZ

Euler Angles

Rotation can also be about either body axis or space axis.

- Body axis
  - local axis
  - relative axis
  - intrinsic axis
- Space axis
  - global axis
  - absolute axis
  - extrinsic axis

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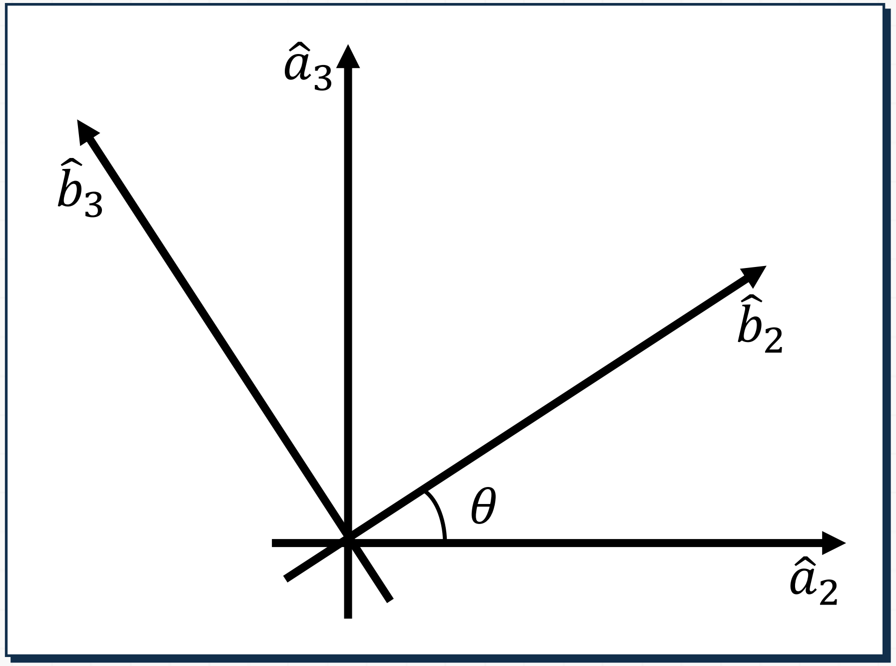
Simple Rotation Matrix

A direction cosine matrix for a rotation about
a simple axis is useful to know, called a rotation matrix.

Simple Rotation Matrix

A rotation matrix about x axis by θ



$$X(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$$

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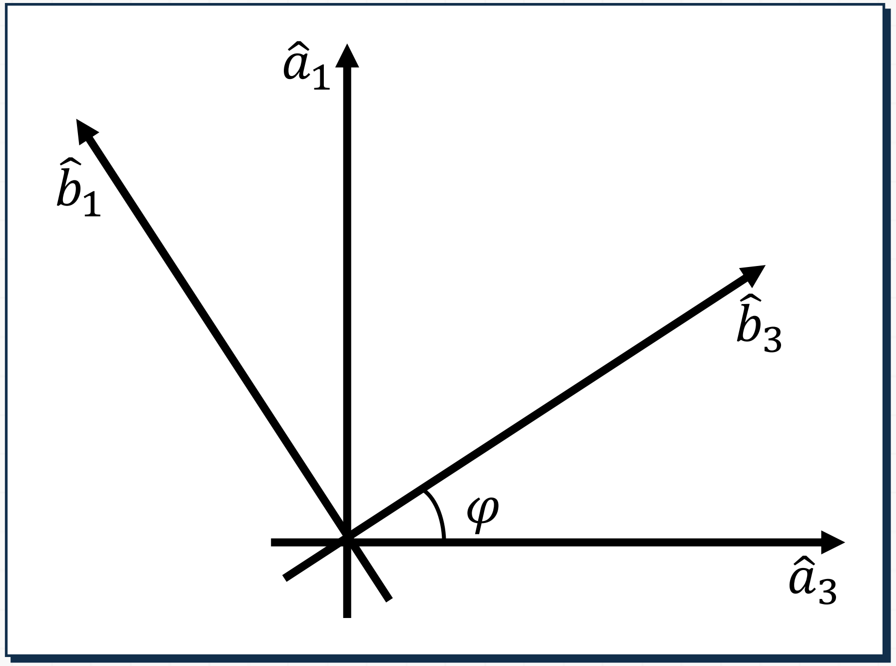
Simple Rotation Matrix

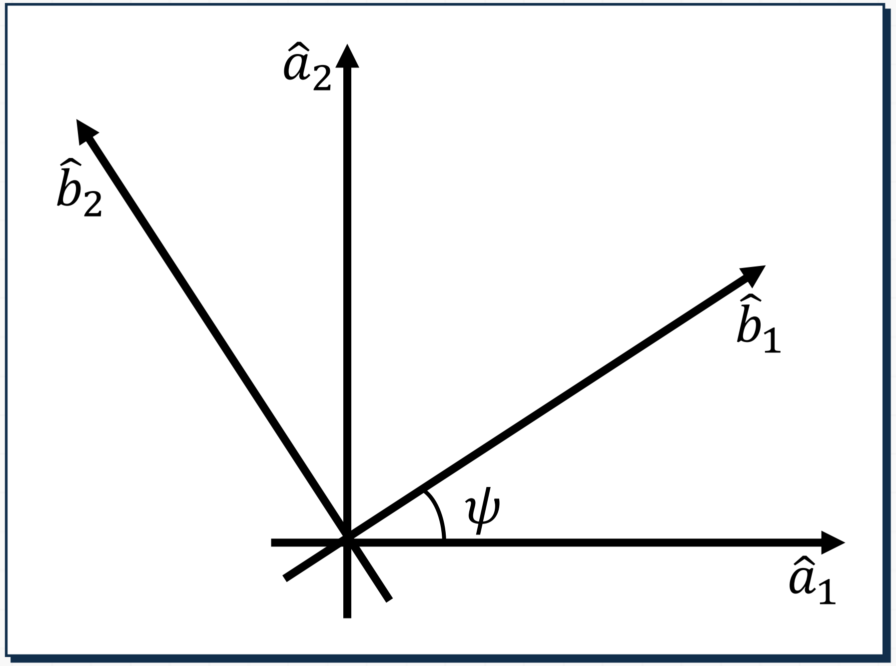
A rotation matrix about y axis by φ



$$Y(\varphi) = \begin{pmatrix} \cos\varphi & 0 & \sin\varphi \\ 0 & 1 & 0 \\ -\sin\varphi & 0 & \cos\varphi \end{pmatrix}$$

Simple Rotation Matrix

A rotation matrix about z axis by ψ



$$Z(\psi) = \begin{pmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Composition of Simple Rotation Matrices

We can easily figure out that any direction cosine matrix can be composed of three consecutive simple rotation matrices.

Composition of Simple Rotation Matrices

Body Axis

- Order of rotation with Euler angles about body axis
  - postmultiplication

- Body axis rotation with XYZ Euler angles

$$DCM = X(\theta)Y(\varphi)Z(\psi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\varphi & 0 & \sin\varphi \\ 0 & 1 & 0 \\ -\sin\varphi & 0 & \cos\varphi \end{pmatrix} \begin{pmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \cos\varphi\cos\psi & -\cos\varphi\sin\psi & \sin\varphi \\ \cos\theta\sin\psi + \sin\theta\sin\varphi\cos\psi & \cos\theta\cos\psi - \sin\theta\sin\varphi\sin\psi & -\sin\theta\cos\psi \\ \sin\theta\sin\psi - \cos\theta\sin\varphi\cos\psi & \sin\theta\cos\psi + \cos\theta\sin\varphi\sin\psi & \cos\theta\cos\psi \end{pmatrix}$$

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Composition of Simple Rotation Matrices

Space Axis

- Order of rotation with Euler angles about space axis
  - premultiplication

- Space axis rotation with XYZ Euler angles

$$DCM = Z(\psi)Y(\varphi)X(\theta) = \begin{pmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos\varphi & 0 & \sin\varphi \\ 0 & 1 & 0 \\ -\sin\varphi & 0 & \cos\varphi \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$$

$$= \begin{pmatrix} \cos\varphi\cos\psi & -\cos\theta\sin\psi + \sin\theta\sin\varphi\cos\psi & \sin\theta\sin\psi + \cos\theta\sin\varphi\cos\psi \\ \cos\varphi\sin\psi & \cos\theta\cos\psi + \sin\theta\sin\varphi\sin\psi & -\sin\theta\cos\psi + \cos\theta\sin\varphi\sin\psi \\ -\sin\varphi & \sin\theta\cos\varphi & \cos\theta\cos\varphi \end{pmatrix}$$

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Properties of Rotation Matrices

Direction Cosine Matrix and Rotation Matrix

- Direction cosine matrix is to describe relative orientation between two coordinate systems.
- Rotation matrix is to rotate a point about an origin to another location in one coordinate system.
- Since the direction cosine matrix and rotation matrix coincide, both terms will be used interchangeably in this course.
- Rotation matrix is a 3x3 orthogonal matrix (i.e., $R^{-1} = R^T$).

Properties of Rotation Matrices

A point (e.g., a position vector) $\vec{p}$

- A point (e.g., a position vector) $\vec{p}$ can be expressed in terms of two coordinate systems ($\hat{a}_1 \hat{a}_2 \hat{a}_3$ and $\hat{b}_1 \hat{b}_2 \hat{b}_3$).

$$\vec{p} = p_{a1}\hat{a}_1 + p_{a2}\hat{a}_2 + p_{a3}\hat{a}_3 = p_{b1}\hat{b}_1 + p_{b2}\hat{b}_2 + p_{b3}\hat{b}_3$$

Properties of Rotation Matrices

A point (e.g., a position vector) $\vec{p}$

- Apply dot product the above equations with $\hat{a}_1 \hat{a}_2 \hat{a}_3$ in a row, then we get the following

$$\begin{pmatrix} p_{a1} \\ p_{a2} \\ p_{a3} \end{pmatrix} = \begin{pmatrix} \hat{a}_1 \cdot \hat{b}_1 & \hat{a}_1 \cdot \hat{b}_2 & \hat{a}_1 \cdot \hat{b}_3 \\ \hat{a}_2 \cdot \hat{b}_1 & \hat{a}_2 \cdot \hat{b}_2 & \hat{a}_2 \cdot \hat{b}_3 \\ \hat{a}_3 \cdot \hat{b}_1 & \hat{a}_3 \cdot \hat{b}_2 & \hat{a}_3 \cdot \hat{b}_3 \end{pmatrix} \begin{pmatrix} p_{b1} \\ p_{b2} \\ p_{b3} \end{pmatrix}$$

- Note that the above expression is not an abuse of notation.

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Properties of Rotation Matrices

A point (e.g., a position vector) $\vec{p}$

- Simply, we have

$$\vec{p}^A = {}^A R^B \vec{p}^B$$

  - where $\vec{p}^A$ is the coordinate vector of the position vector $\vec{p}$ in terms of coordinate system $\hat{a}_1\hat{a}_2\hat{a}_3$
  - $\vec{p}^B$ is the coordinate vector of the position vector $\vec{p}$ in terms of coordinate system $\hat{b}_1\hat{b}_2\hat{b}_3$
  - ${}^A R^B$ is the rotation matrix of coordinate system $\hat{b}_1\hat{b}_2\hat{b}_3$ with respect to coordinate system $\hat{a}_1\hat{a}_2\hat{a}_3$

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Properties of Rotation Matrices

Actual action of rotating a point to another

- A rotation matrix rotates one point to another location within the same coordinate system.
  - For example, a point (1,0,0) can be rotated by $90°$ about z axis as follows.

$$Z(90°)\begin{pmatrix}1\\0\\0\end{pmatrix}=\begin{pmatrix}\cos 90° & -\sin 90° & 0\\ \sin 90° & \cos 90° & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}1\\0\\0\end{pmatrix}=\begin{pmatrix}0\\1\\0\end{pmatrix}$$

Properties of Rotation Matrices

Actual action of rotating a point to another

- Simply, we have

$$\vec{p}^{\,\text{rotated}} = R\vec{p}^{\,\text{original}}$$

Properties of Rotation Matrices

Actual action of rotating a point to another

- Note that the actual action of rotating a point to another should be done within one coordinate system.
- Also, note that $\vec{p}^A = {}^A R^B \vec{p}^B$ relates two coordinate systems to a point.

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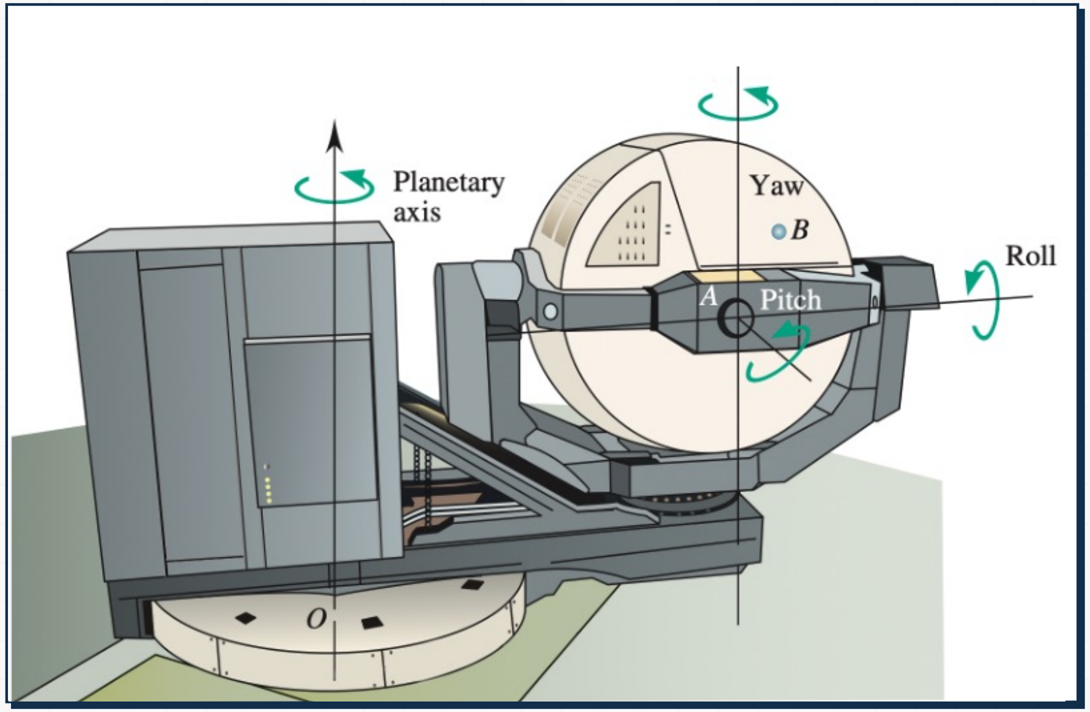
Sample Problem 1

Consider the following flight simulator.



- Determine
  - The orientation of the cab with respect to control cube

Sample Problem 1

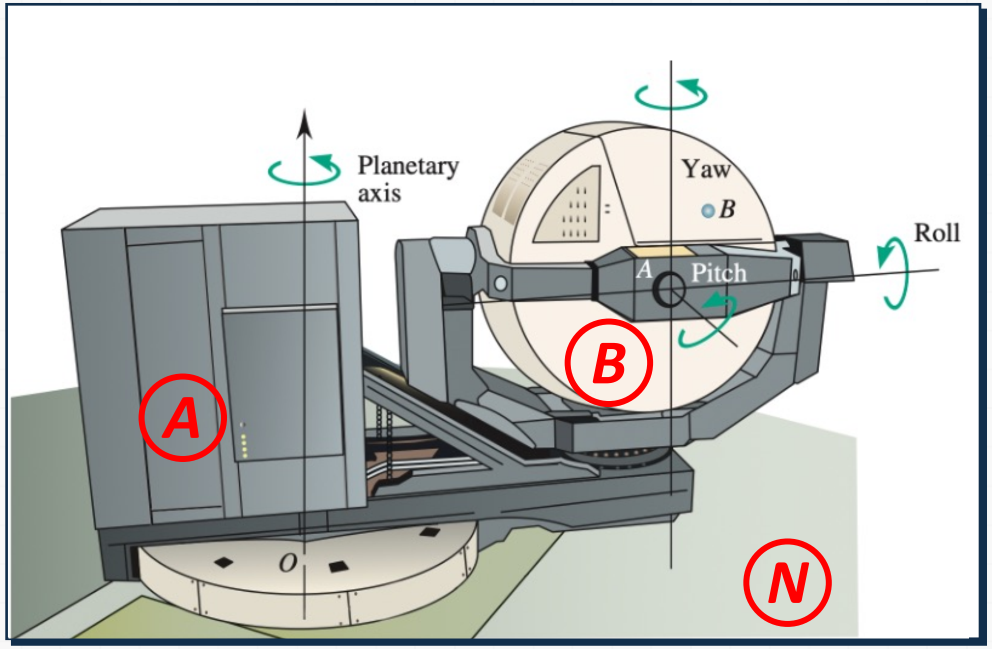
Consider the following flight simulator.

- Strategy
  - Identify appropriate reference frames.
  - Assign appropriate coordinate systems to each reference frame.
  - Build the direction cosine table by performing necessary dot products.

Sample Problem 1

Modeling and Analysis

- Assign a coordinate system to each reference frame



- Each rigid body has one reference frame

Inertial reference frame: N

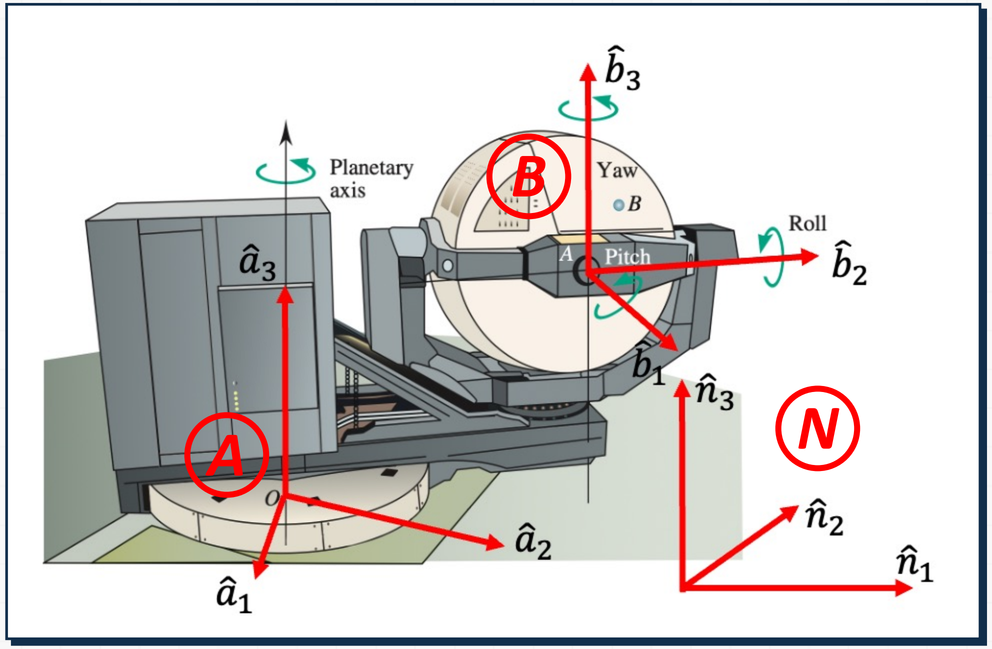
Reference frame of cube: A

Reference frame of cab: B

Sample Problem 1

Modeling and Analysis

- Identify reference frames



- Each reference frame has at least one coordinate system

Inertial reference frame: N
Assign $\hat{n}_1\hat{n}_2\hat{n}_3$

Reference frame of cube: A
Assign $\hat{a}_1\hat{a}_2\hat{a}_3$

Reference frame of cab: B
Assign $\hat{b}_1\hat{b}_2\hat{b}_3$

Sample Problem 1

Modeling and Analysis

- Gather information for coordinate systems
  - Assume that we are given the following information.

$$
\begin{aligned}
\hat{a}_1 &= 0.866\hat{n}_1 + 0.5\hat{n}_2 \\
\hat{a}_2 &= -0.5\hat{n}_1 + 0.866\hat{n}_2 \\
\hat{a}_3 &= \hat{n}_3
\end{aligned}
$$

$$
\begin{aligned}
\hat{b}_1 &= 0.75\hat{n}_1 + 0.433\hat{n}_2 - 0.5\hat{n}_3 \\
\hat{b}_2 &= -0.217\hat{n}_1 + 0.875\hat{n}_2 + 0.433\hat{n}_3 \\
\hat{b}_3 &= 0.625\hat{n}_1 - 0.217\hat{n}_2 + 0.75\hat{n}_3
\end{aligned}
$$

Sample Problem 1

Modeling and Analysis

- Complete the direction cosine table
  - Compute dot products

$$
\begin{array}{c|ccc}
 & \hat{b}_1 & \hat{b}_2 & \hat{b}_3 \\
\hline
\hat{a}_1 & \hat{a}_1\cdot\hat{b}_1 & \hat{a}_1\cdot\hat{b}_2 & \hat{a}_1\cdot\hat{b}_3 \\
\hat{a}_2 & \hat{a}_2\cdot\hat{b}_1 & \hat{a}_2\cdot\hat{b}_2 & \hat{a}_2\cdot\hat{b}_3 \\
\hat{a}_3 & \hat{a}_3\cdot\hat{b}_1 & \hat{a}_3\cdot\hat{b}_2 & \hat{a}_3\cdot\hat{b}_3
\end{array}
=
\begin{array}{c|ccc}
 & \hat{b}_1 & \hat{b}_2 & \hat{b}_3 \\
\hline
\hat{a}_1 & 0.866 & 0.25 & 0.433 \\
\hat{a}_2 & 0 & 0.866 & -0.5 \\
\hat{a}_3 & -0.5 & 0.433 & 0.75
\end{array}
$$

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Sample Problem 1

Reflect and Think

Identifying the right reference frames and defining appropriate coordinate systems are the key.

Usually, the relation between coordinate systems are given as consecutive rotations with Euler angles.

Once direction cosine matrix is found, rotation matrix is found as well.

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Sample Problem 2

Perform consecutive rotations.

- Determine
  - Rotation matrix to rotate about z axis by $30°$, followed by a rotation about the new y axis by $30°$, followed by a rotation about the new x axis by $30°$
  - Rotation matrix to rotate about z axis by $30°$, followed by a rotation about the original y axis by $30°$, followed by a rotation about the original x axis by $30°$

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Sample Problem 2

Perform consecutive rotations.

- Strategy
  - List the order of rotation with the Euler angles.
  - Identify the type of axis of rotation: body axis vs. space axis.

Body axis rotation → postmultiplication

Space axis rotation → premultiplication

  - Construct the rotation matrix with the right order and right direction.

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Sample Problem 2

Perform consecutive rotations.

- Determine
  - Rotation matrix to rotate about z axis by 30°, followed by a rotation about the new y axis by 30°, followed by a rotation about the new x axis by 30° → zyx with body axis rotations (postmultiplication)

$$
\begin{aligned}
R &= Z(30°)Y(30°)X(30°) \\
&= \begin{pmatrix} \cos 30° & -\sin 30° & 0 \\ \sin 30° & \cos 30° & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 30° & 0 & \sin 30° \\ 0 & 1 & 0 \\ -\sin 30° & 0 & \cos 30° \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos 30° & -\sin 30° \\ 0 & \sin 30° & \cos 30° \end{pmatrix} \\
&= \begin{pmatrix} 0.75 & -0.217 & 0.625 \\ 0.433 & 0.875 & -0.217 \\ -0.5 & 0.433 & 0.75 \end{pmatrix}
\end{aligned}
$$

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Sample Problem 2

Perform consecutive rotations.

Determine

- Rotation matrix to rotate about z axis by $30°$, followed by a rotation about the original y axis by $30°$, followed by a rotation about the original x axis by $30°$
$\rightarrow zyx$ with space axis rotations (premultiplication)

$$
\begin{aligned}
R &= X(30°)Y(30°)Z(30°) \\
&= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos 30° & -\sin 30° \\ 0 & \sin 30° & \cos 30° \end{pmatrix} \begin{pmatrix} \cos 30° & 0 & \sin 30° \\ 0 & 1 & 0 \\ -\sin 30° & 0 & \cos 30° \end{pmatrix} \begin{pmatrix} \cos 30° & -\sin 30° & 0 \\ \sin 30° & \cos 30° & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 0.75 & -0.433 & 0.5 \\ 0.650 & 0.625 & -0.433 \\ -0.125 & 0.650 & 0.75 \end{pmatrix}
\end{aligned}
$$

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Sample Problem 2

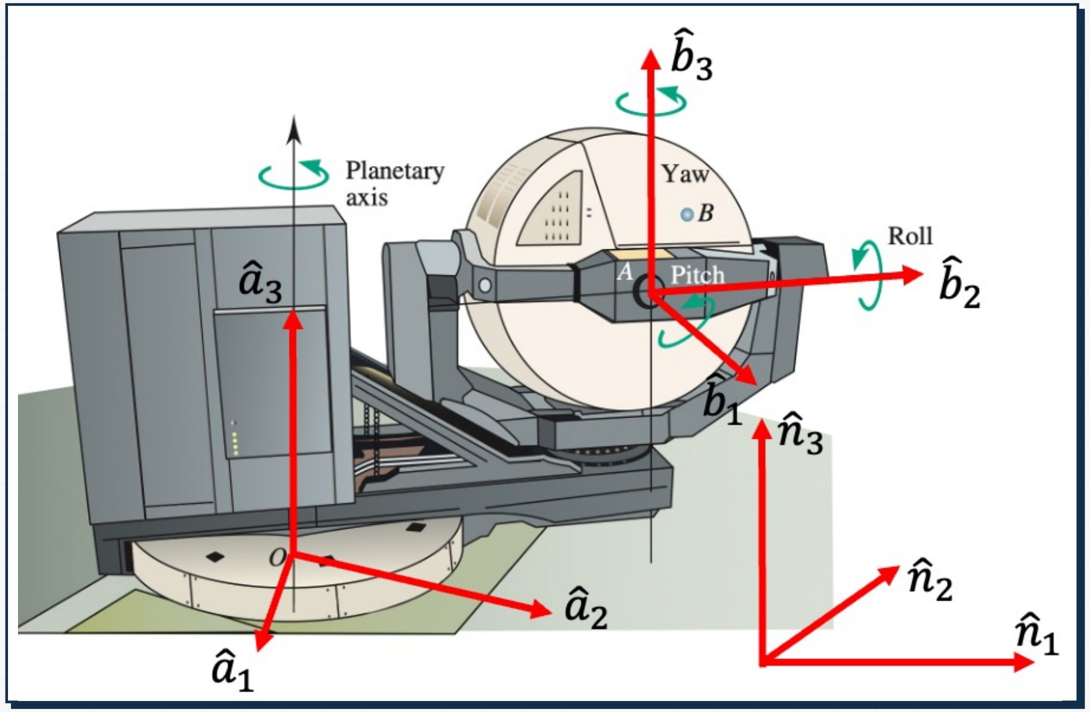
Reflect and Think

Even though the order of rotations with Euler angles are the same, different types of axis of rotation (i.e., body axis vs. space axis) will result in different rotation matrix.

The order of rotations with body axis results in the same rotation matrix if the order is reversed in the space axis.

Sample Problem 3

Consider the following flight simulator.



- Determine
  - The coordinate vector of $\overrightarrow{AB}$ in terms of $\hat{b}_1\hat{b}_2\hat{b}_3$ if

$$\overrightarrow{AB} = 0.683\hat{a}_1 + 0.366\hat{a}_2 + 1.183\hat{a}_3$$

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Sample Problem 3

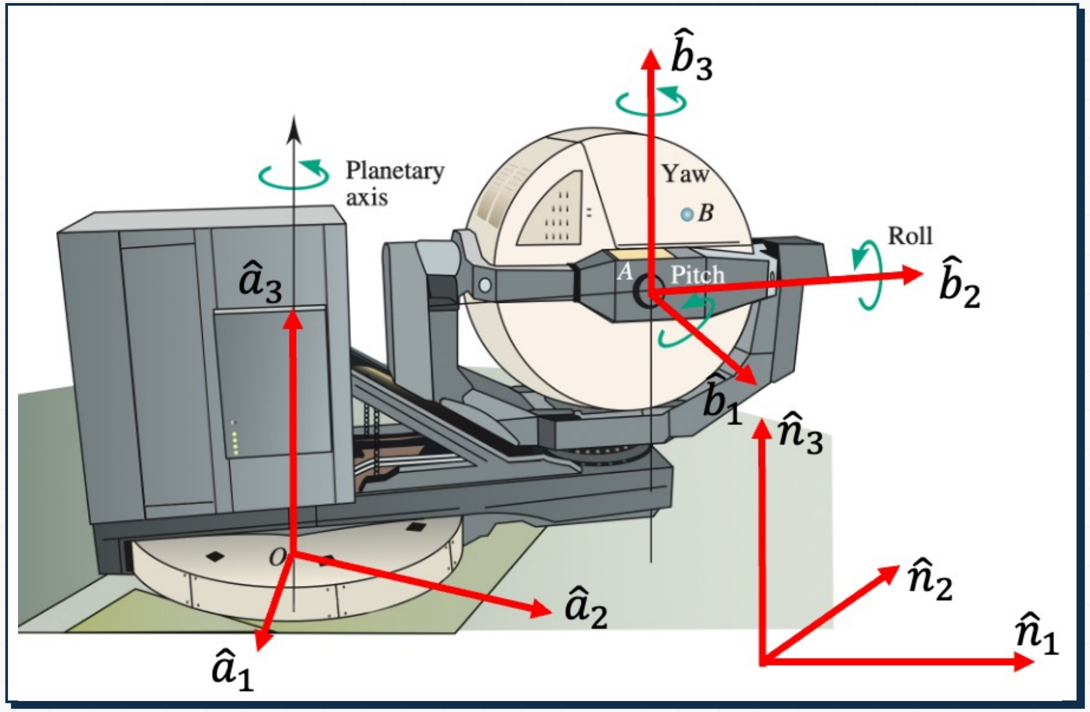
Consider the following flight simulator.

- Strategy
  - Find the rotation matrix between $\hat{a}_1\hat{a}_2\hat{a}_3$ and $\hat{b}_1\hat{b}_2\hat{b}_3$.
  - Consider the formula that relates two different coordinate vectors of a same position vector using two different coordinate systems.
  - Inverse of rotation matrix is transpose of itself.

Sample Problem 3

Modeling and Analysis

- Consider the following flight simulator.



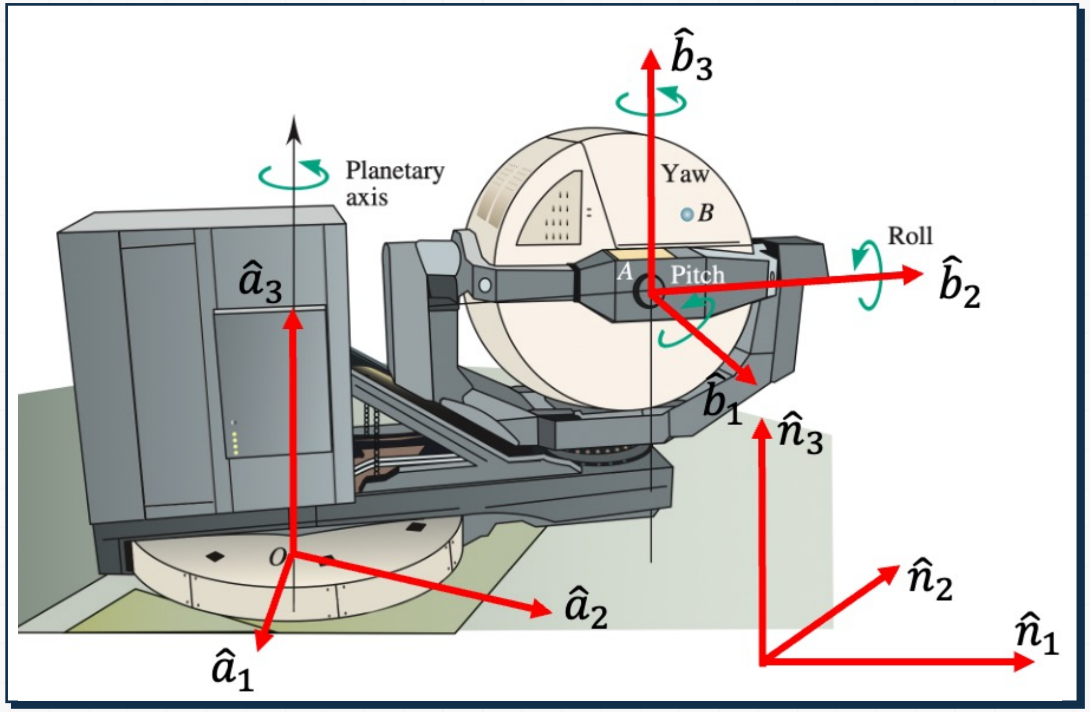
- Direction cosine matrix that maps $\hat{a}_1\hat{a}_2\hat{a}_3$ to $\hat{b}_1\hat{b}_2\hat{b}_3$ was computed in the Sample problem 1.

$$ {}^{A}R^{B} = \begin{pmatrix} 0.866 & 0.25 & 0.433 \\ 0 & 0.866 & -0.5 \\ -0.5 & 0.433 & 0.75 \end{pmatrix} $$

Sample Problem 3

Modeling and Analysis

- Consider the following flight simulator.



Consider the following formula.

$$\overrightarrow{AB}^A = {}^A R^B \, \overrightarrow{AB}^B$$

Since we need to find $\overrightarrow{AB}^B$, try

$$\overrightarrow{AB}^B = \left({}^A R^B\right)^T \overrightarrow{AB}^A$$

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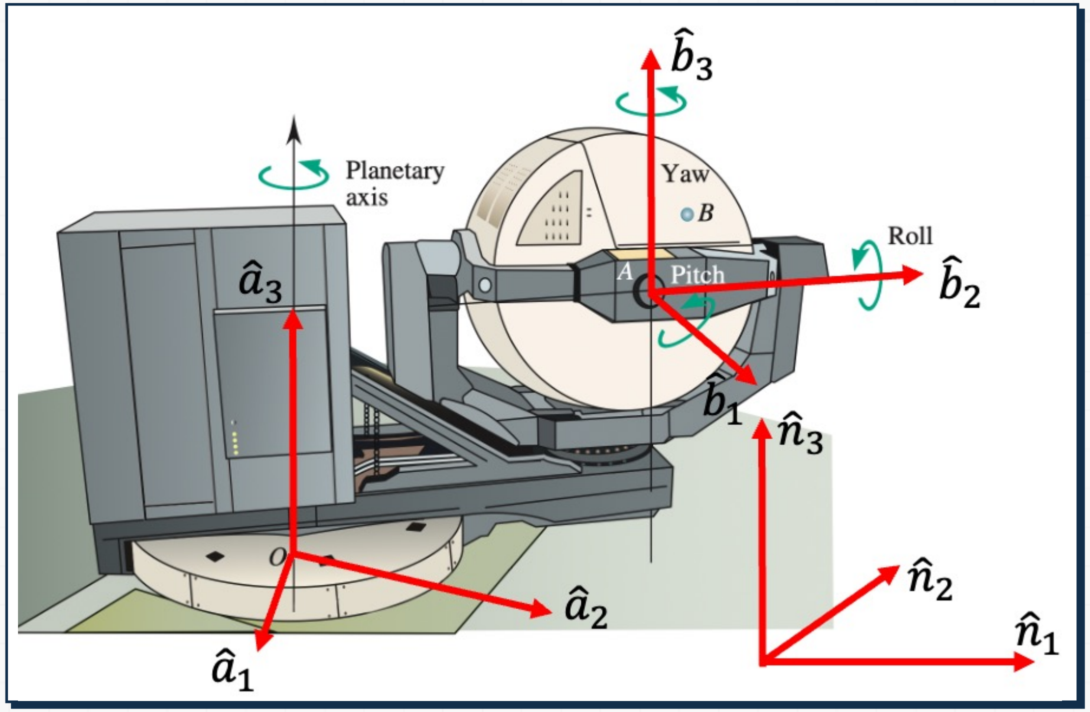
Sample Problem 3

Modeling and Analysis

- Consider the following flight simulator.



$$\overrightarrow{AB}^B = \left({}^A R^B\right)^T \overrightarrow{AB}^A$$
$$= \begin{pmatrix} 0.866 & 0.25 & 0.433 \\ 0 & 0.866 & -0.5 \\ -0.5 & 0.433 & 0.75 \end{pmatrix}^T \begin{pmatrix} 0.683 \\ 0.366 \\ 1.183 \end{pmatrix}$$
$$= \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

Therefore,

$$\overrightarrow{AB} = 0.683\hat{a}_1 + 0.366\hat{a}_2 + 1.183\hat{a}_3$$
$$= \hat{b}_2 + \hat{b}_3$$

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Sample Problem 3

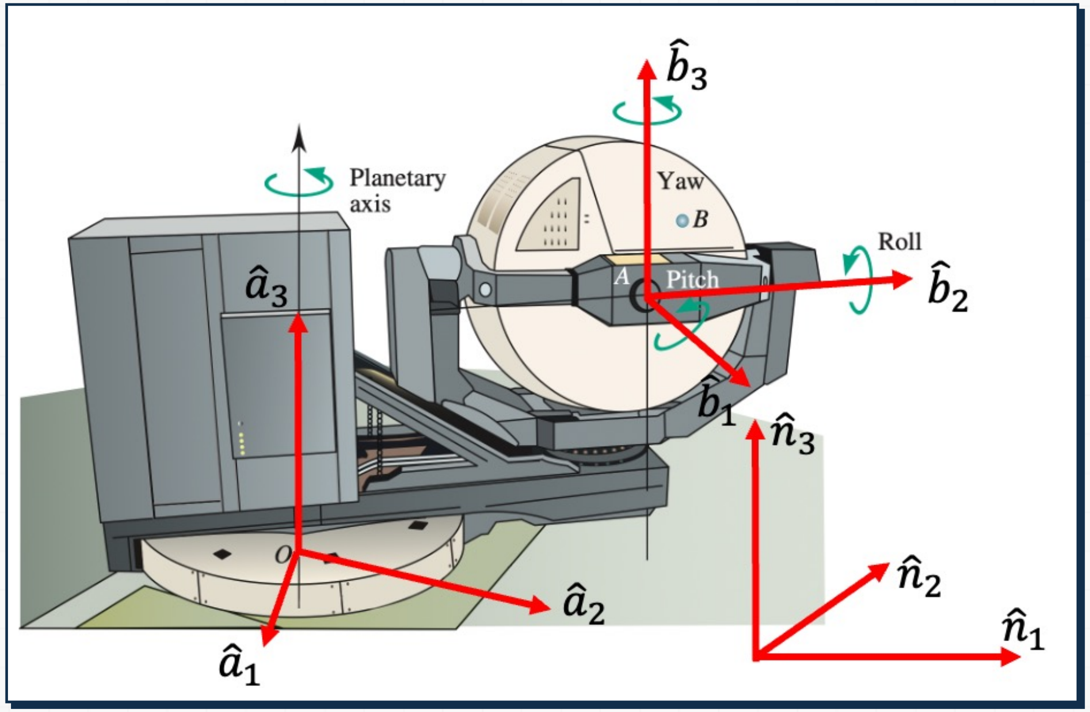
Reflect and Think

Direction cosine matrix is identical to the rotation matrix.
When $\overrightarrow{AB}$ is expressed in terms of $\hat{a}_1\hat{a}_2\hat{a}_3$ and $\hat{b}_1\hat{b}_2\hat{b}_3$, origins of the coordinate systems do not matter.
Only the directions of $\hat{a}_1\hat{a}_2\hat{a}_3$ and $\hat{b}_1\hat{b}_2\hat{b}_3$ are used.

Sample Problem 4

Consider the following flight simulator.

- From the image below, the cab rotates $45°$ more in pitch direction.



- Determine
  - The coordinate vector of $\overrightarrow{AB}$ in terms of $\hat{b}_1\hat{b}_2\hat{b}_3$.

Sample Problem 4

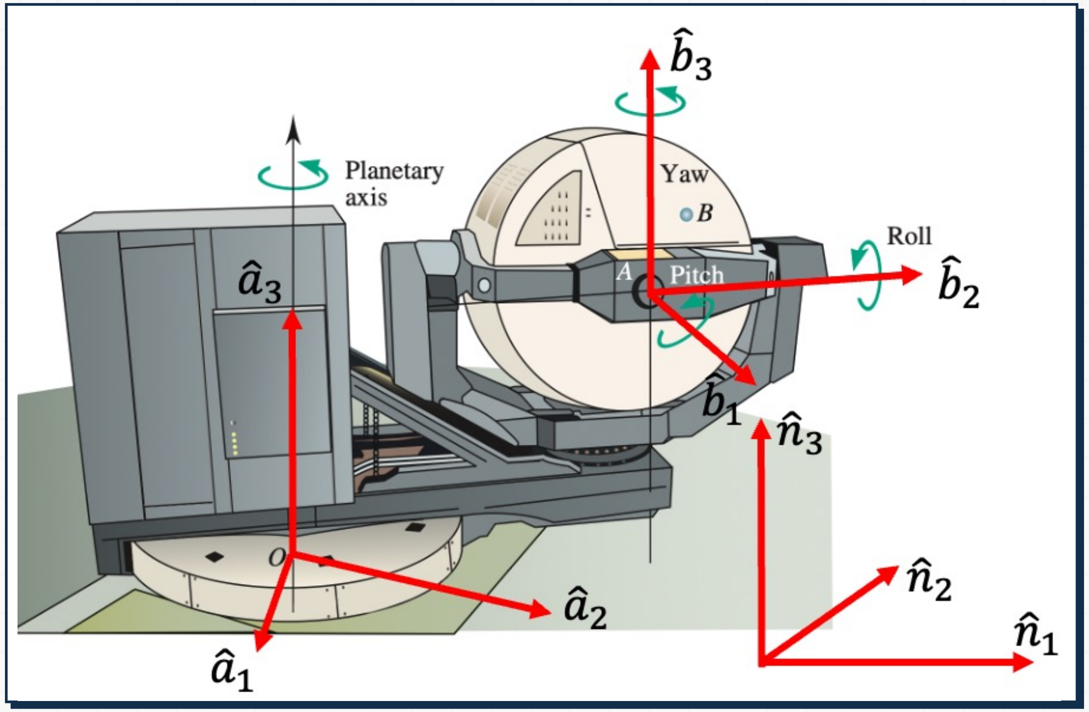
Consider the following flight simulator.

- Strategy
  - Find the rotation matrix for $45°$ in x direction.
  - Rotate the original point using the rotation matrix.

Sample Problem 4

Modeling and Analysis

- Find the rotation matrix for $45°$ in x direction.



From the Sample problem 3, we have

$$\overrightarrow{AB}^B = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

The rotation matrix for $45°$ in x direction is

$$X(45°) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos 45° & -\sin 45° \\ 0 & \sin 45° & \cos 45° \end{pmatrix}$$

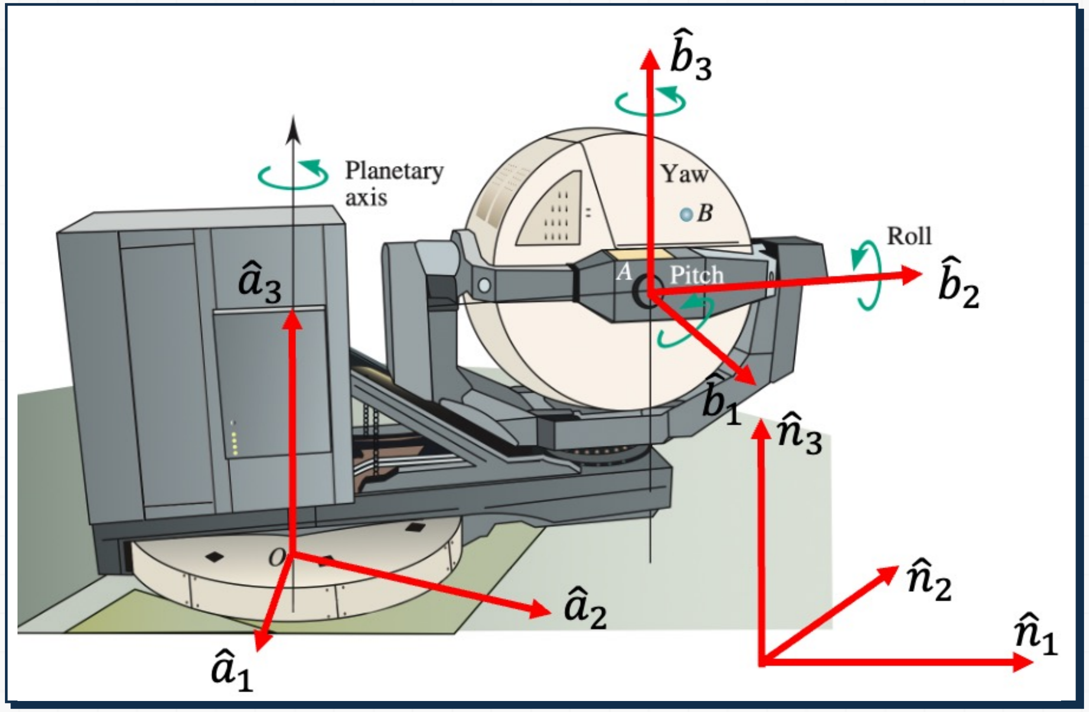
$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0.707 & -0.707 \\ 0 & 0.707 & 0.707 \end{pmatrix}$$

GIST

Sample Problem 4

Modeling and Analysis

- Rotate the point.



From the Sample problem 3, we have

$$\overrightarrow{AB}^{rotated} = X(45°)\begin{pmatrix}0\\1\\1\end{pmatrix}$$

$$= \begin{pmatrix}1 & 0 & 0\\0 & 0.707 & -0.707\\0 & 0.707 & 0.707\end{pmatrix}\begin{pmatrix}0\\1\\1\end{pmatrix}$$

$$= \begin{pmatrix}0\\0\\1.414\end{pmatrix}$$

Therefore,

$$\overrightarrow{AB}^{rotated} = 1.414\hat{b}_3$$

GIST

Sample Problem 4

Reflect and Think

When a rotation matrix is used to rotate a point, both the original point and the rotated point are expressed in terms of the same coordinate system.

References

- Beer, Johnston, Cornwell, Self, Sanghi, "Vector Mechanics for Engineers: Dynamics" 12th Edition, McGraw Hill, Chapter 15
- 유홍희, "동역학", 1판, 문운당, 8장